4.2 The sources, origins, physical and working properties of thermoforming and thermosetting polymers and their social and ecological footprint	To apply knowledge and understanding of the advantages, disadvantages and applications of the following materials, in order to be able to discriminate between them and select appropriately.
	4.2.1 Thermoforming polymers: a acrylic (in topic 1) b high impact polystyrene (HIPS) (in topic 1) c biodegradable polymers – Biopol® (in topic 1) d polystyrene (PS) rigid (high density polystyrene) and expanded polystyrene e Styrofoam™ f polyvinyl chloride g acrylonitrile-butadiene-styrene (ABS) h polyethylene terephthalate (PET) i urethane/polyurethane j fluoroelastomer.
	4.2.2 Thermosetting polymers: a polyester resin (in topic 1) (when used to reinforce glass and carbon fibre matting) b urea formaldehyde (in topic 1).
	4.2.3 Sources and origins – where thermoforming and thermosetting polymers are resourced/manufactured and their geographical origin: a Russia, UAE, Saudi Arabia – crude oil.
	4.2.4 The physical characteristics of each polymer: a density b durability.
	4.2.5 Working properties – the way in which each material behaves or responds to external sources: a insulator of heat (in topic 1) b insulator of electricity (in topic 1) c toughness (in topic 1) d plasticity e hardness f tensile strength g compressive strength.

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Key idea	What students need to learn:
	4.2.6 Social footprint: a trend forecasting b impact of extraction and material production on the environment c impact of extraction and material production on wildlife d ease and difficulty of recycling and disposal.
	4.2.7 Ecological footprint: a sustainability b oil exploration and extraction c wildlife loss d processing e transportation f wastage g pollution.